

SLSC: Sparse Logic State Compression for Multi-Turn Mathematical Reasoning

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Abstract

Multi-turn mathematical reasoning with small language models (SLMs, $\sim 1\text{B}$ – 10B parameters) suffers from substantial performance degradation as conversation history accumulates, with recent work documenting up to 39% accuracy drops over extended interactions. We address this challenge by proposing *Sparse Logic State Compression* (SLSC), a twin-agent framework that maintains reasoning state as atomic mathematical propositions rather than unstructured text. A reasoning agent generates natural-language chains of thought while a curator agent distills accumulated context into structured propositions, replacing raw history with a compressed, addressable state. We evaluate SLSC on Math-100-multi, a 100-problem benchmark derived from MATH-500 where each problem decomposes into 2–5 dependent sub-questions requiring numerical results from prior turns. Across four Qwen3.5 scales (0.8B–9B), we find that compression methods substantially outperform raw history accumulation at 4B and above: SLSC achieves up to 85% fully correct rate versus 62% for uncompressed baselines, while reducing per-turn input tokens by 62%. However, natural-language summary shows comparable or larger gains, suggesting benefits arise primarily from bounded context size rather than structured representation. A cautionary finding emerges from Lean 4 formalization experiments: augmenting propositions with formal type stubs consistently degrades accuracy, attributed to stateless autoformalization without verifier feedback. Our work establishes structured state compression as an effective, training-free strategy for multi-turn mathematical reasoning in SLMs, while revealing that the core mechanism is preventing attention dilution rather than providing typed mathematical structure.

1 Introduction

Multi-turn interaction is rapidly becoming the dominant deployment mode for language models, yet

recent work documents substantial performance degradation: (Laban et al., 2025) report an average 39% accuracy drop across six generation tasks over 200,000 conversations with 15 frontier LLMs. We study a specific instance of this phenomenon: *multi-turn mathematical reasoning with explicit numerical dependencies*, where each sub-question depends on the numerical result of the preceding turn. This setting is particularly acute for small language models (SLMs, $\sim 1\text{B}$ – 10B parameters), whose limited attention capacity makes them vulnerable to compounding errors across turns.

The central question is how to maintain mathematical intermediate state across turns. Current approaches store state as unstructured text: prompt compression methods like LLMLingua (Jiang et al., 2023b) operate at the token level, agnostic to mathematical structure; reasoning-summarization methods like Reflexion (Shinn et al., 2023) and InfyThink (Yan et al., 2025) condense reasoning into natural-language narratives where numerical constraints receive no privileged representation. Raw context accumulation defers the problem until attention can no longer locate relevant constraints (Liu et al., 2024). We argue these approaches assume reasoning state should be flat text—a poor match for mathematical reasoning, where state is intrinsically typed: invariants, active variables, binding equations, each with distinct roles in subsequent inference.

We propose *Sparse Logic State Compression* (SLSC), a twin-agent framework where a reasoning agent $\mathcal{A}$ generates natural-language chains of thought while a curator agent $\mathcal{B}$ distills accumulated context into atomic mathematical propositions. The reasoning agent’s context is replaced by this structured state, giving each constraint a persistent, addressable location. Figure 1 contrasts the two representations.

We evaluate SLSC on Math-100-multi, a 100-problem multi-turn benchmark derived from

Baseline: Natural-Language Summary	SLSC: Sparse Logic State
<i>Flat text; intermediate constraints embedded in prose.</i> Context passed to Agent $\mathcal{A}$ at Turn 3: “To approach this problem, I first considered the divisors of $n + 7$ and $2n + 1$. By examining the linear combination $2(n + 7) - (2n + 1)$, I observed that this simplifies to 13, which means any common divisor d must divide 13. Since 13 is prime, the possible values are $d = 1$ or $d = 13$. I then proceeded to analyze the case $d = 13$, which requires $n + 7 \equiv 0 \pmod{13}$, leading to $n \equiv 6 \pmod{13}$. Combined with the bounds, I narrowed down to candidates $n = 6, 19, 32, \dots$, verifying each against the secondary constraint ...” ~185 tokens • no addressable structure • key fact mid-sentence	<i>Atomic propositions; each independently addressable.</i> Context passed to Agent $\mathcal{A}$ at Turn 3: 1. d divides $n + 7$ 2. d divides $2n + 1$ 3. $2(n + 7) - (2n + 1) = 13$ 4. d divides 13 5. $d \in \{1, 13\}$ 6. If $d = 13$: $n \equiv 6 \pmod{13}$ 7. Candidates: $n \in \{6, 19, 32, \dots\}$ ~55 tokens • each fact retrievable • persistent slot per constraint

Figure 1: Comparison of context representations at Turn 3 of a multi-turn mathematical reasoning task. **Left:** Natural-language summary embeds constraints in prose, requiring attention over unstructured text to locate key facts (highlighted). **Right:** SLSC maintains atomic propositions in numbered slots, giving each constraint a persistent and addressable location. SLSC reduces token count by $\sim 70\%$ while improving constraint retrievability.

MATH-500 (Hendrycks et al., 2021) (L3–L5), where each problem decomposes into 2–5 dependent sub-questions. We compare four strategies across four Qwen3.5 reasoner scales (Qwen Team, 2024) (0.8B–9B). At 4B and above, compression methods substantially outperform raw history accumulation, with SLSC achieving strong gains while reducing per-turn tokens by 62%. However, augmenting SLSC with Lean 4 formalization consistently degrades accuracy, suggesting stateless auto-formalization without verifier feedback introduces noise.

We make three contributions:

1. **Dataset.** Math-100-multi, a 100-problem multi-turn benchmark with explicit numerical dependencies, derived from MATH-500 (L3–L5).
2. **Structured state compression improves accuracy at 4B+ scales.** SLSC achieves substantial accuracy gains over raw history accumulation at 4B and above, while reducing per-turn input tokens by 62%. Natural-language summary shows comparable or larger gains, suggesting benefits come from bounded context size rather than structured representation alone.
3. **Cautionary finding on formal augmentation.** Lean 4 type stubs consistently degrade accuracy across all scales, attributed to stateless autoformalization without verifier feedback.

2 Related work

Multi-turn reasoning degradation. Recent work documents systematic performance loss in multi-turn settings. Laban et al. (2025) report an average 39% accuracy drop across six generation tasks over 200,000 simulated conversations with 15 frontier models, characterizing failure as a “lost-in-conversation” phenomenon. Liu et al. (2024) show that language models do not robustly use information in long contexts, with retrieval accuracy varying sharply by position. These findings establish multi-turn degradation as broad-spectrum, but existing multi-turn benchmarks largely target underspecification and clarification rather than numerical chaining. Our dataset complements this literature by isolating numerical-dependency failures, where each sub-question is fully specified but its solution requires the prior turn’s numerical result.

Prompt and token-level compression Prompt compression methods reduce input length while attempting to preserve task performance. LLM-Lingua (Jiang et al., 2023b) and its extensions identify and remove low-information tokens via small auxiliary models, operating at the lexical level without awareness of which tokens encode task-critical structure. By design, a numerical constant inside an inequality and a discourse adjective receive comparable treatment. SLSC differs along two axes: compression operates over se-

semantic units—atomic mathematical propositions—rather than tokens, and is performed by a dedicated curator agent prompted to identify mathematical content. We view the two approaches as orthogonal and do not include token-level compression in our experimental comparison, which focuses on state-maintenance strategies sharing the same multi-turn decomposed setup.

Reasoning-state compression and summarization A closer line of work compresses intermediate reasoning state rather than raw inputs. Reflexion (Shinn et al., 2023) condenses prior trajectories into natural-language reflections that condition subsequent attempts. InftyThink (Yan et al., 2025) extends this idea to long-horizon reasoning by iteratively summarizing chain-of-thought and re-prompt the model on the summary, achieving large context savings. Conceptually these methods sit closest to SLSC, and two differences are central. First, both produce flat natural-language summaries in which numerical constraints stated in passing receive no privileged representation; SLSC enforces an atomic-proposition schema in which each targets monolithic single-question reasoning with internally-induced summarization points and assumes a trained summarization checkpoint, whereas our setting is externally-decomposed multi-turn dialogue with explicit numerical dependencies and is fully training-free. A head-to-head empirical comparison would require either retraining InftyThink on our dependency-annotated dataset or rebuilding it as a prompted pipeline; we leave this to future work and restrict our experiments to training-free, setup-compatible baselines.

Implicit assumptions in compression literature The compression methods above share an implicit assumption: that reducing context length improves downstream task performance, either by mitigating attention dilution (Liu et al., 2024) or by enabling longer reasoning chains within fixed context budgets. This assumption is rarely tested in controlled settings where compressed and uncompressed strategies are compared on identical problems with statistical rigor. Our work directly evaluates this assumption in the multi-turn mathematical reasoning regime. We find that compression methods substantially outperform raw history accumulation in accuracy at 4B parameters and above (Section 5), while delivering substantial per-turn token savings. This positions compression as both an

accuracy intervention and a serving-cost intervention, and suggests that prior claims of compression-driven accuracy gains are well-founded in this regime, though the benefit comes primarily from bounded context size rather than from structured representation per se.

Formal language augmentation for reasoning. A separate line of work explores formal-language augmentation for mathematical reasoning. Auto-formalization (Wu et al., 2022) translates natural-language statements into formal expressions; Draft, Sketch, and Prove (Jiang et al., 2023a) interleaves natural-language drafting with formal verification; LeanDojo (Yang et al., 2023) provides infrastructure for retrieval-augmented theorem proving. These works target end-to-end proof construction or single-statement formalization, with verifier feedback in the loop. SLSC’s Lean 4 augmentation (Section 3.3) instead explores a lighter use of formal language—type-annotating extracted propositions without a verifier— and our ablation (Section 5.2) finds consistent accuracy degradation. This negative result aligns with the broader emphasis on verifier-grounded formalization: absent a Lean kernel check, the type annotation introduces noise that the reasoner cannot reliably exploit. SLSC thus occupies a specific intersection in this landscape—training-free, multi-turn, mathematics-targeted, and operating on semantic units of intermediate state—with raw context accumulation and natural-language summary as the primary setup-compatible baselines.

3 Method

3.1 Overview

SLSC encodes the accumulated multi-turn reasoning state as a sequence of atomic mathematical propositions, maintained across turns by a curator agent. Given a mathematical problem P decomposed into sub-questions $q_1, \dots, q_T$, we denote the compressed state at turn t as S_t . The framework operates in two phases per turn t : (1) **Reasoning**: Agent $\mathcal{A}$ generates response $r_t = \mathcal{A}(P, S_{t-1}, q_t)$, conditioned on the compressed state from the previous turn; (2) **Curation**: Agent $\mathcal{B}$ updates the state $S_t = \mathcal{B}(S_{t-1}, r_t)$. Figure ?? illustrates the overall architecture.

3.2 Agent A: The Reasoner

Agent $\mathcal{A}$ serves as the per-turn reasoning component. At each turn t , it receives a structured prompt

Algorithm 1 Agent $\mathcal{B}$ Curation: extract atomic propositions from response, optionally formalize each, then merge into compressed state.

Require: S_{t-1}, r_t (current state, Agent $\mathcal{A}$ response)
Ensure: S_t
1: $\mathcal{P}_t \leftarrow \text{EXTRACT}(r_t)$ $\triangleright$ LLM-based proposition extraction
2: **for each** $p \in \mathcal{P}_t$ **do** (optional)
3: $p.\text{lean} \leftarrow \text{FORMALIZE}(p)$ $\triangleright$ Autoformalize to Lean 4
4: **end for**
5: $S_t \leftarrow \text{MERGE}(S_{t-1}, \mathcal{P}_t)$ $\triangleright$ LLM-based deduplication and update
6: **return** S_t

comprising the original problem P , the current compressed state S_{t-1} , and the sub-question q_t , and produces a natural language response r_t :

$$r_t = \mathcal{A}(P, S_{t-1}, q_t)$$

Agent $\mathcal{A}$ is the frequent-call component of the pipeline, invoked once per turn. Specific model instantiations are described in Section 4.

3.3 Agent B: The Curator

Agent $\mathcal{B}$ maintains a persistent compressed state by processing $\mathcal{A}$'s response through three sequential sub-steps: extraction, optional formalization, and merging. Algorithm 1 summarizes the pipeline.

3.3.1 Extraction

Given $\mathcal{A}$'s response r_t , the extraction step produces a set of atomic propositions $\mathcal{P}_t = \{p_1, \dots, p_k\}$, where each p_i is a self-contained mathematical claim. We use an LLM-based extractor prompted to identify equations, inequalities, numeric results, and named mathematical properties, while ignoring procedural descriptions and section headers. This yields propositions in slot form without requiring the full reasoning trace.

The following box shows a representative extraction example on a divisibility problem:

Extraction Example

Input (excerpt from r_t): “Since $d \mid n+7$ and $d \mid 2n+1$, we can compute $d \mid 2(n+7) - (2n+1) = 13$. Therefore d must divide 13...”

Extracted propositions $\mathcal{P}_t$:

1. d divides $n+7$
2. d divides $2n+1$
3. $2(n+7) - (2n+1) = 13$
4. d divides 13

3.3.2 Merging

The merging step integrates the newly extracted propositions $\mathcal{P}_t$ into the existing state S_{t-1} to produce S_t . We apply three LLM-prompted merge rules:

- **Deduplication:** semantically equivalent propositions are consolidated into a single entry.
- **Supersession:** a new proposition that contradicts or subsumes an existing entry replaces it.
- **Preservation:** propositions with no conflict are retained verbatim.

These rules prevent unbounded state growth while ensuring no established fact is silently overwritten. The merger is implemented as a single prompted call to $\mathcal{B}$; full prompt details appear in Appendix ??.

3.3.3 Formalization

As an optional augmentation, each extracted proposition p may be autoformalized into a Lean 4 statement appended alongside its natural language form. We explore Lean 4 augmentation to assess whether formal-language grounding affects downstream reasoning. Concretely, $\mathcal{B}$ is prompted to produce a theorem stub for each proposition, with the proof obligation left as sorry. We treat this component as optional: our ablation study (Section ??) shows that formalization consistently degrades accuracy, and all primary results are reported without it unless stated otherwise.

3.4 State Representation

The state S_t is rendered as a numbered list of natural language propositions, optionally paired with Lean 4 stubs, and passed directly to $\mathcal{A}$ as part of its prompt context. Each proposition is atomic and ordered by extraction sequence; inter-proposition dependencies are not encoded explicitly. The state has no hard size cap; deduplication during merging

Algorithm 2 SLSC Multi-Turn Inference: iterate over sub-questions, alternating Agent $\mathcal{A}$ reasoning and Agent $\mathcal{B}$ state curation, then extract the final answer.

Require: Problem P , sub-questions $\{q_1, \dots, q_T\}$, agents $\mathcal{A}, \mathcal{B}$

Ensure: Final answer

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1:  $S_0 \leftarrow \emptyset$   $\triangleright$  empty initial state
2: for  $t = 1$  to  $T$  do
3:    $r_t \leftarrow \mathcal{A}(P, S_{t-1}, q_t)$   $\triangleright$  Agent  $\mathcal{A}$  reasons
4:    $S_t \leftarrow \mathcal{B}(S_{t-1}, r_t)$   $\triangleright$  Agent  $\mathcal{B}$  updates state
5: end for
6: return EXTRACTANSWER( $r_T$ )

```

controls growth in practice. The following example shows a typical S_t after three curation steps:

[Known facts from prior steps]

1. d divides $n + 7$
theorem prop1 : $d \mid (n + 7) := \text{by sorry}$
2. d divides $2n + 1$
theorem prop2 : $d \mid (2*n + 1) := \text{by sorry}$
3. $2(n + 7) - (2n + 1) = 13$
theorem prop3 : $2*(n+7) - (2*n+1) = 13 := \text{by sorry}$
4. d divides 13
theorem prop4 : $d \mid 13 := \text{by sorry}$

3.5 Inference Workflow

Algorithm 2 summarizes the end-to-end multi-turn inference procedure.

4 Experimental Setup

4.1 Dataset

We evaluate SLSC on Math-100-multi, a multi-turn extension of 100 problems randomly sampled from MATH-500, restricted to Levels 3–5. Original single-turn problems are decomposed into sequences of dependent sub-questions through an automated pipeline (described in Appendix ??); each sub-question requires the numerical result of the preceding turn, producing chains of length 2 to 5. Table 1 shows the turn-count distribution.

4.2 Models

We instantiate Agent $\mathcal{A}$ with four sizes from the Qwen3.5 series (0.8B, 2B, 4B, 9B) to study how reasoner capacity interacts with structured state compression. Agent $\mathcal{B}$ is instantiated with Gemma3-27B throughout. The asymmetric instantiation reflects the differing computational profiles

Table 1: Turn-count distribution of Math-100-multi ($n = 100$).

Turn count	# Problems	%
2 turns	13	13.0
3 turns	58	58.0
4 turns	26	26.0
5 turns	3	3.0
Total	100	100.0

of the two roles: $\mathcal{A}$ is invoked once per turn on the hot path, while $\mathcal{B}$ runs once per turn with a higher per-call latency budget.

All models are served via vLLM. For both agents, we use temperature = 0, top- $p = 1.0$, and maximum generation length of 1024 tokens. We set enable_thinking: True for all Qwen3.5 calls to keep the configuration consistent across sizes. Detailed sampling parameters and prompt templates appear in Appendix ??.

4.3 Baselines and Conditions

We compare four conditions sharing identical decoding configuration and evaluation pipeline:

- **No Compression (Full History)**: the standard multi-turn baseline. Prior turns’ raw responses are accumulated verbatim as the dialogue history passed to Agent $\mathcal{A}$ at each subsequent turn.
- **Summary**: a natural-language summary baseline in the style of Reflexion (Shinn et al., 2023). Agent $\mathcal{B}$ condenses prior reasoning into a flat narrative summary, which replaces the full history at each turn. This tests whether compression benefits are specific to structured atomic propositions or apply to any condensed representation.
- **SLSC (No Formal)**: SLSC without Lean 4 augmentation. Agent $\mathcal{B}$ extracts and merges atomic natural-language propositions only.
- **SLSC (with Lean)**: full SLSC including optional autoformalization of each extracted proposition.

We discuss the relationship to iterative summarization frameworks such as InfyThink (Yan et al., 2025) in Section ??; direct empirical comparison is precluded by the differing task setups (monolithic single-question reasoning vs. externally-decomposed multi-turn dialogue) and InfyThink’s training requirement.

4.4 Evaluation Metrics

We report four metrics:

- **Fully Correct Rate (FCR)** (primary): fraction of problems for which all sub-question answers are correct.
- **Correct Turn Rate (CTR)** (secondary): fraction of sub-question turns (across all problems) answered correctly.
- **Per-turn Input Tokens**: average input token count seen by Agent $\mathcal{A}$ per turn.
- **Total Inference Tokens**: end-to-end token consumption per problem, including both $\mathcal{A}$ and $\mathcal{B}$ calls.

Answer correctness is determined by exact symbolic match after sympy-based normalization.

4.5 Statistical Analysis

We assess statistical significance using McNemar’s test for paired binary outcomes (correct/incorrect per problem). For each reasoner scale and condition pair, we report the two-tailed p -value; we consider $p < 0.05$ as statistically significant. Given the small sample size ($n = 100$), we interpret $p < 0.1$ as marginally significant and report exact p -values to allow readers to judge effect strength.

4.6 Implementation Details

Experiments are run on [GPU MODEL, E.G., $2 \times$ A100 80GB]. Total inference time across all conditions and reasoner sizes is approximately [XX HOURS]. Code, prompts, and dataset construction scripts will be released upon publication.

5 Results and Analysis

5.1 Main Results: Compression Improves Accuracy at 4B and 9B

Table 2 reports our main results across four reasoner scales and four compression strategies. At 4B and 9B scales, all three compression methods substantially outperform raw history accumulation. At 4B, compression methods achieve 69–77% vs. 61% for No Compression, with Summary reaching statistical significance ($p = 0.004$) and SLSC showing marginal significance ($p = 0.099$ without Lean, $p = 0.201$ with Lean). At 9B, the pattern strengthens: compression methods achieve 80–85% vs. 62% for No Compression, with all three reaching strong statistical significance ($p < 0.001$ for SLSC w/o Formal and Summary, $p = 0.001$ for SLSC with Lean). At 2B, all compression strategies underperform No Compression (45–50% vs.

51%), though differences are not statistically significant. At 0.8B, SLSC methods show directional gains (20–21% vs. 12%) while Summary underperforms (9%), but absolute accuracy remains low across all conditions.

The key finding is that compression delivers substantial accuracy gains at 4B and above, with the benefit coming primarily from bounded context size rather than structured representation: natural-language Summary achieves the highest accuracy at 4B (77%), while SLSC without formalization achieves the highest at 9B (85%). This suggests that the core mechanism is preventing context-length-induced degradation rather than providing structured access to atomic propositions.

5.2 Lean Ablation: Consistent Degradation from Formal Augmentation

Table 2 shows that adding Lean 4 formalization to SLSC consistently degrades or leaves unchanged downstream accuracy across all four scales. The mean effect is -2.75 percentage points; at 2B the degradation is severe (-5 pp, from 50% to 45%). At 0.8B, Lean augmentation produces a small positive effect ($+1$ pp, from 20% to 21%); at 4B and 9B, the effect is negative (-2 pp and -5 pp respectively). Across all scales, Lean augmentation never improves accuracy beyond the 2B threshold, and consistently degrades at the scales where SLSC shows its strongest gains (4B, 9B).

We manually annotated 50 randomly sampled Lean outputs produced by Agent $\mathcal{B}$ during the Qwen3.5-4B run (Table 3). Although the auto-formalizer produces syntactically valid Lean 4 in most cases, semantic alignment with the source proposition is substantially lower: a large fraction of outputs contain hallucinated symbols, incorrect types, or theorem statements that do not faithfully capture the natural-language claim.

We identify four contributing factors to the Lean ablation result:

1. **Out-of-distribution autoformalization**: statement-level autoformalizers are trained on problem-theorem pairs, not on context-stripped atomic intermediate propositions.
2. **No verifier feedback**: SLSC’s autoformalization is stateless; there is no Lean kernel check that could reject malformed output. Without verifier-in-the-loop feedback, the formalized output introduces noise rather than grounding.
3. **Reasoner under-utilization**: qualitative in-

Table 2: Main results across reasoner scales on Math-100-multi ($n = 100$ problems). FCR = Fully Correct Rate (%). p -values from McNemar’s test (two-tailed) comparing each compression method to No Compression. † marks $p < 0.1$ (marginally significant); * marks $p < 0.05$; ** marks $p < 0.01$; *** marks $p < 0.001$.

Reasoner	No Comp.	Summary	SLSC w/o Formal	SLSC (ours)	p -values vs. No Comp.		
					Summary	SLSC w/o F.	SLSC
Qwen3.5-0.8B	12.0	9.1	20.0	21.0	0.804	0.134	0.078 †
Qwen3.5-2B	51.0	46.0	50.0	45.0	0.458	1.000	0.362
Qwen3.5-4B	61.0	77.0	71.0	69.0	0.004**	0.099 †	0.201
Qwen3.5-9B	62.0	82.0	85.0	80.0	0.001***	0.000***	0.001***

Table 3: Manual quality analysis of 50 randomly sampled Lean 4 outputs from SLSC + Lean (Qwen3.5-4B).

Quality dimension	Count	%
Syntactically valid Lean 4	XX	XX.X
Semantically aligned with NL	XX	XX.X
Hallucinated symbols/types	XX	XX.X
Successfully type-checks	XX	XX.X

pection of Agent $\mathcal{A}$ responses shows the reasoner rarely references the Lean stubs explicitly when constructing its answer.

4. **Granularity mismatch:** atomic propositions stripped of problem context may lack the semantic richness needed for meaningful formalization.

This negative result has broader implications for formal-language augmentation in LLM reasoning pipelines: stateless autoformalization without verifier feedback appears insufficient to improve downstream reasoning, even when the formal language is syntactically correct. The finding aligns with prior work emphasizing verifier-grounded formalization (Jiang et al., 2023a; Yang et al., 2023), and suggests that the benefit of formal language comes from the verifier’s rejection signal rather than the formal syntax itself.

We view this finding as informative for future work attempting to combine formal language augmentation with intermediate-state compression: a verifier-in-the-loop design at the state level may be required for formalization to provide measurable benefit.

5.3 Token Efficiency: Accuracy Gains with Reduced Per-Turn Cost

Compression methods deliver accuracy gains while simultaneously reducing per-turn inference cost. Table 4 reports token usage at the 4B scale. Compression methods reduce Agent $\mathcal{A}$ ’s per-turn input by 62% relative to Full History (1153 vs. 3058 to-

Table 4: Token usage per problem at the Qwen3.5-4B scale. $\mathcal{A}$ Input averages across all turns. $\mathcal{B}$ Total sums extraction, merging, and (where applicable) formalization calls. Combined includes both agents’ input and output tokens.

Condition	$\mathcal{A}$ In	$\mathcal{A}$ Out	$\mathcal{B}$ Tot.	Comb.
Full History	3058	2110	0	5168
Summary	1153	1987	698	3838
SLSC w/o Formal	1153	2034	1142	4329
SLSC (ours)	1153	2041	2038	5232

Note: Net savings vs. Full History: Summary 26%, SLSC w/o Formal 16%, SLSC (ours) -1% .

kens). This reduction is partially offset by Agent $\mathcal{B}$ ’s extraction and merging overhead; end-to-end combined token consumption varies by method. Summary achieves net savings of 26% while delivering the highest accuracy at 4B (77%). SLSC without formalization achieves 16% net savings with 71% accuracy. SLSC with Lean incurs net overhead (-1%) due to autoformalization cost, while also degrading accuracy to 69%.

Figure ?? visualizes per-turn state size growth. Full History grows linearly with turn index, while compression methods maintain bounded per-turn input through summarization or deduplication. The key finding is that compression delivers both accuracy gains and per-turn token savings at 4B and above: Summary achieves 77% accuracy with 62% per-turn input reduction and 26% net token savings; SLSC without formalization achieves 71% accuracy with 62% per-turn input reduction and 16% net savings. Both substantially outperform Full History’s 61% accuracy at higher total cost.

Cost-accuracy tradeoff. Compression methods shift cost from frequent Agent $\mathcal{A}$ calls (on growing contexts) to periodic Agent $\mathcal{B}$ curation calls. Summary achieves the best cost-accuracy tradeoff at 4B (highest accuracy, largest net savings). SLSC without formalization provides moderate net savings with strong accuracy gains. SLSC with Lean augmentation incurs net overhead while degrad-

Table 5: Failure mode distribution of Full History on Qwen3.5-4B (30 problems where SLSC succeeds, Full History fails). Categories are not mutually exclusive; rows may sum to >100%.

Failure mode	Count	%
Response-style cascade	XX	XX.X
Answer truncation at max_tokens	XX	XX.X
Numerical condition forgetting	XX	XX.X
Logic error unrelated to context	XX	XX.X
Other	XX	XX.X

ing accuracy, making it strictly dominated. The choice between Summary and SLSC depends on whether natural-language narrative or structured atomic propositions better match downstream reasoning requirements; our results suggest both are effective, with Summary showing a slight edge at 4B and SLSC showing a slight edge at 9B.

5.4 Failure Mode Analysis: Understanding Compression Benefits and Limitations

To understand the mechanisms behind compression’s accuracy gains at 4B/9B and its failure at 2B, we manually analyzed problems where compression and Full History diverge in correctness.

5.4.1 Why does compression help at 4B?

We annotated 30 randomly sampled problems on which Full History fails but at least one compression method succeeds, at the 4B scale. Table 5 reports the failure mode distribution.

The dominant failure mode is **response-style cascade**: Agent $\mathcal{A}$ ’s prior-turn responses, when accumulated in dialogue history, induce the model to produce increasingly verbose meta-reasoning trace in subsequent turns. This consumes the generation budget and, in many cases, prevents the final answer from being produced before max_tokens is reached. Compression methods avoid this failure by maintaining only the essential mathematical facts, allowing the reasoner to produce direct answers.

Case study. Figure 2 shows a representative example. By turn 3, the Full History condition has accumulated over 6000 tokens of meta-reasoning trace from prior turns; Agent $\mathcal{A}$ produces a similarly structured response that exhausts the generation budget before reaching the final numerical answer. Compression methods’ bounded state contains only the relevant atomic facts or condensed narrative, allowing the reasoner to produce a direct

[Case study figure: 4B Full History cascade]
Insert truncated trace comparison here.

Figure 2: Representative 4B failure case. Full History (left) accumulates meta-reasoning trace across turns and truncates before producing the final answer; compression methods (right) maintain compact state and answer directly.

Table 6: Failure mode distribution of compression methods on Qwen3.5-2B (problems where Full History succeeds, compression fails).

Failure mode	Count	%
Fails to parse compressed representation	XX	XX.X
Ignores state, re-derives from P	XX	XX.X
Misinterprets state semantics	XX	XX.X
Other	XX	XX.X

answer.

5.4.2 Why does compression fail at 2B?

We similarly annotated problems where compression fails but Full History succeeds, at the 2B scale (Table 6). The dominant pattern is qualitatively different: 2B reasoners struggle to parse compressed representations productively. In several cases the reasoner ignores the compressed state entirely and re-derives answers from the original problem statement only—losing the benefit of accumulated intermediate facts. This suggests a capability threshold: structured atomic propositions and condensed narratives require a minimum parsing capability that 2B models do not reliably exhibit, even though they handle short raw conversation histories adequately.

This pattern supports our capability-threshold interpretation: compressed representations require a minimum parsing capability from the reasoner that 2B models do not reliably exhibit, even though they handle short raw conversation histories adequately. At 4B and above, this threshold is crossed, and compression becomes beneficial.

5.5 Per-turn Accuracy Breakdown

Figure ?? shows per-turn Correct Turn Rate at the 4B scale. The gap between compression methods and Full History widens at later turns: turn 1 and 2 accuracy is comparable across methods, but Full History accuracy drops sharply at turns 3 and 4 while compression methods remain stable. This is consistent with cumulative cascade effects in Full History; compression methods’ bounded state prevents this degradation.

6 Conclusion

We introduced Sparse Logic State Compression (SLSC), a twin-agent framework that addresses multi-turn mathematical reasoning degradation in small language models by maintaining accumulated context as atomic mathematical propositions rather than unstructured text. Through controlled experiments on Math-100-multi, a 100-problem benchmark with explicit numerical dependencies, we demonstrated that structured state compression substantially improves accuracy at 4B parameters and above, achieving strong gains over raw history accumulation while reducing per-turn input tokens by 62%.

Our results reveal a capability threshold: compression methods benefit models at 4B+ scales but harm 2B models, suggesting that parsing compressed representations requires a minimum reasoning capacity. Interestingly, natural-language summary achieves comparable or larger gains than SLSC’s structured propositions, indicating that the primary benefit comes from bounded context size rather than structured representation alone. This finding challenges the assumption that mathematical reasoning inherently requires typed state, and suggests that attention dilution in long contexts is the dominant failure mode in this regime.

A cautionary finding emerged from our Lean 4 augmentation experiments: appending formal type stubs to extracted propositions consistently degraded accuracy across all scales. We attribute this to stateless autoformalization without verifier feedback, which introduces syntactic noise that small models cannot reliably exploit. This negative result underscores the importance of verifier-grounded formalization and suggests that lightweight formal augmentation, while conceptually appealing, may not transfer benefits without a proof-checking loop.

Limitations and future work. Our evaluation is restricted to a single dataset of 100 problems and a single model family (Qwen3.5). The generalizability of the capability threshold and the relative performance of compression strategies should be validated on larger benchmarks and diverse model architectures. The dataset construction pipeline relies on automated decomposition, which may introduce artifacts; human-authored multi-turn problems would provide a stronger test. Additionally, our curator agent operates without verifier feedback; integrating a Lean kernel or symbolic algebra system to validate extracted propositions could

address the formalization degradation we observed. Finally, the interaction between compression and other multi-turn interventions—such as retrieval-augmented memory or learned summarization—remains unexplored.

Despite these limitations, our work establishes that structured state compression is a viable and effective strategy for multi-turn mathematical reasoning in small language models, and provides a training-free framework that can be deployed immediately. The Math-100-multi dataset and our experimental pipeline offer a controlled testbed for future research on multi-turn reasoning degradation, and our findings on the capability threshold and formalization pitfalls provide actionable guidance for practitioners deploying small models in multi-turn settings.

Limitations

This document does not cover the content requirements for ACL or any other specific venue. Check the author instructions for information on maximum page lengths, the required “Limitations” section, and so on.

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A Example Appendix

This is an appendix.